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import json
import os
from datetime import datetime

FILE_NAME = "expenses.json"

def load_expenses():
    if os.path.exists(FILE_NAME):
        with open(FILE_NAME, "r") as f:
            return json.load(f)
    return []

def save_expenses(expenses):
    with open(FILE_NAME, "w") as f:
        json.dump(expenses, f, indent=4)

def add_expense(expenses):
    title = input("Enter expense title: ")
    amount = float(input("Enter amount: "))
    category = input("Enter category: ")
    date = datetime.now().strftime("%Y-%m-%d")
    expenses.append({
        "id": len(expenses) + 1,
        "title": title,
        "amount": amount,
        "category": category,
        "date": date
    })
    save_expenses(expenses)
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        print("Expense added successfully.\n")

def view_expenses(expenses):
    if not expenses:
        print("No expenses recorded.\n")
        return
    print(f"{'ID':<5}{'Date':<12}{'Title':<20}{'Category':<15}{'Amount':<10}")
    print("-" * 62)
    for e in expenses:
        print(f"{e['id']:<5}{e['date']:<12}{e['title']:<20}{e['category']:<15}{e['amount']:<10.2f}")
    print()

def delete_expense(expenses):
    view_expenses(expenses)
    if not expenses:
        return
    try:
        exp_id = int(input("Enter ID of expense to delete: "))
    except ValueError:
        print("Invalid ID.\n")
        return
    new_expenses = [e for e in expenses if e["id"] != exp_id]
    if len(new_expenses) == len(expenses):
        print("Expense not found.\n")
        return
    for i, e in enumerate(new_expenses, start=1):
        e["id"] = i
    save_expenses(new_expenses)

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        expenses.clear()
        expenses.extend(new_expenses)
        print("Expense deleted
successfully.\n")

def total_by_category(expenses):
    if not expenses:
        print("No expenses recorded.\n")
        return
    totals = {}
    for e in expenses:
        totals[e["category"]] =
totals.get(e["category"], 0) + e["amount"]
    print(f"'Category':<20}'Total':<10}")
    print("-" * 30)
    for cat, amt in totals.items():
        print(f"{cat:<20}{amt:<10.2f}")
    print(f"\nGrand Total:
{sum(totals.values()):.2f}\n")

def main():
    expenses = load_expenses()
    while True:
        print("==== Expense Tracker
====")
        print("1. Add Expense")
        print("2. View Expenses")
        print("3. Delete Expense")
        print("4. View Totals by Category")
        print("5. Exit")
        choice = input("Enter your choice:
")

        if choice == "1":

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        add_expense(expenses)
    elif choice == "2":
        view_expenses(expenses)
    elif choice == "3":
        delete_expense(expenses)
    elif choice == "4":
        total_by_category(expenses)
    elif choice == "5":
        print("Goodbye!")
        break
    else:
        print("Invalid choice. Try
again.\n")

if __name__ == "__main__":
    main()
```